The F = 1 target-in-walk arm at k = 28 and 32, single-target ecbench (2026-11-08)

Class. Measured, single-target ecbench sessions under a protocol frozen before data (PROTOCOL.md hashes in `data/frozen.sha256`, freeze commit `e6a759cc72`). Counted units only (ecbench GAE, S = cost/sqrt(r)); every comparison at w = 0, 0.1 and 1 under both band readings: the pre-2026-11-08 reading (table inserts unpriced; this README's earlier "committed band") and the priced reading (`ecbench band --price-table-work`, the default since the owner ruling of 2026-11-08, landing in PR #3726). Public synthetic Curve25519-family miniatures; no security claim. No pcc claim has been filed; every number is an unverified assertion until it is reviewed [pcc:unverified]. No code change; binary `/home/user/ctarget-f1/release/ecbench` (SHA-256 `d23fab99...`, `data/binary-k{28,32}.sha256`).

Question. The target-in-walk arm had been measured at k = 20 and 24 only. Does its distance to rho-neg hold near 1.4x at k = 28 and 32, where rho-neg's set-up overhead vanishes and both arms sit near their walk floors, and do 32 or 64 jumps pay there, where the set-up they cost is 0.01 sqrt(r) or less?

Answer. The distance holds. The 16-jump arm costs 1.395x [1.345, 1.446] rho-neg at k = 28 and 1.366x [1.267, 1.473] at k = 32 at w = 0, and its lower end stays above 1 at every weight under the priced reading (table inserts priced like rho's canonicalisations); under the pre-2026-11-08 reading the verdict against rho-neg is unresolved in all 16 sessions and the order flips at w = 1 [pcc:unverified]. Against the table-then-probe F = 1 arm it costs 0.906x [0.877, 0.935] and 0.857x [0.811, 0.905] [pcc:unverified]. Its mean S at k = 32 is 1.387 with a walk of 1.368 sqrt(r), 3 % above the second-collision floor of 1.329; rho-neg's search is 0.999 sqrt(r) there, 13 % above its first-collision floor of 0.886, which is why the measured ratio is 1.37x (frozen model 1.38x at k = 28, 1.375x at k = 32; data within 1.1 %) and not the floor ratio 1.5 [pcc:unverified]. More jumps do not pay: 32 jumps give 0.997x [0.954, 1.041] at k = 28 and 0.985x [0.930, 1.044] at k = 32, 64 jumps 0.998x and 0.990x, all intervals containing 1 [pcc:unverified]. Six of seven frozen predictions held; the only failure is the risky P4b (32 jumps strictly below 16 at k = 32) [pcc:unverified]. "Rho-neg is ahead at every weight" holds under the priced reading only [pcc:unverified].

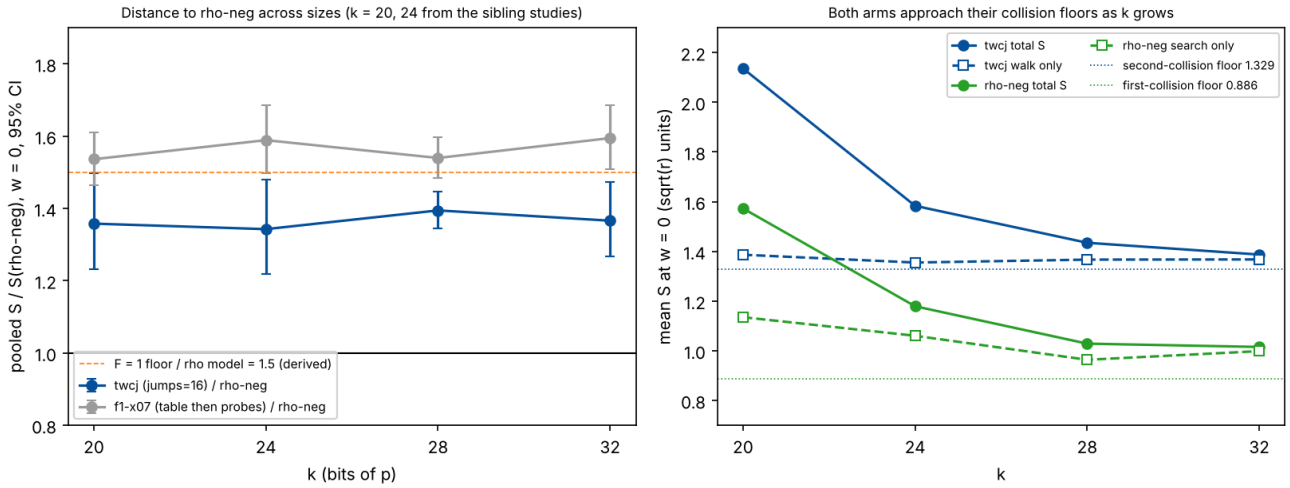


Figure 1: Scaling of the distance to rho

1. Design (as frozen)

8 sessions per size, `targets_per_curve = 1`, 64 measured rounds plus 1 warm-up per arm, `order = alternate`; arms `rho-neg`, `f1-x07` (baseline, `wwalk:x=0.7,g=1,grow=2,seed=run`), `twcj-j16` (primary), `twcj-j32`, `twcj-j64` (`wwalk:x=1.5,g=1,grow=2,tw=1,cj=1,jumps=J,seed=run`), all pipeline arms on `prime-projected-abscissa:size=1,scan_cap=32` with `targets=stride-generator`. Curves: the family miniatures of the autolab holdouts at k = 28 (`r = 33,557,809`) and k = 32 (`r = 536,884,639`). Audit: `ecbench verify --replay 1` per session. Pooling as in the sibling studies. Runtime: 50 s at k = 28, 76 s at k = 32 (`data/run-*.log`).

2. Results (measured)

All 16 sessions completed with 325 of 325 records verified each; every audit reproduced its replay with 0 problems (`data/verify-k*-s*.txt`) [pcc:unverified]. Pooled figures use the 2,560 measured records per size (`data/pooled_measured.json`).

Mean S per arm (pre-2026-11-08 reading, inserts unpriced; the last two columns price the table inserts at w):

k	arm	S w=0	S w=0.1 (unpriced)	S w=1 (unpriced)	w=0.1, inserts priced	w=1, inserts priced
28	rho-neg	1.029	1.125	1.993	1.125	1.993
28	f1-x07	1.584	1.639	2.131	1.735	3.100
28	twcj-j16	1.435	1.435	1.438	1.572	2.805

28	twcj-j32	1.430	1.430	1.433	1.565	2.784
28	twcj-j64	1.432	1.433	1.436	1.566	2.765
32	rho-neg	1.015	1.115	2.014	1.115	2.014
32	f1-x07	1.619	1.677	2.201	1.779	3.218
32	twcj-j16	1.387	1.387	1.388	1.524	2.756
32	twcj-j32	1.367	1.367	1.368	1.501	2.711
32	twcj-j64	1.373	1.373	1.374	1.507	2.717

Pooled ratios with 95 % intervals over 8 single-target sessions:

k	ratio	w = 0	w = 0.1 (pre-2026-11-08 / inserts priced)	w = 1 (pre-2026-11-08 / inserts priced)
28	twcj-j16 / rho-neg	**1.395** [1.345, 1.446]	1.275 [1.230, 1.322] / 1.397 [1.347, 1.448]	0.722 [0.695, 0.749] * / 1.407 [1.356, 1.461]
28	f1-x07 / rho-neg	1.540 [1.484, 1.598]	1.456 / 1.542	1.070 * / 1.556
28	twcj-j16 / f1-x07	**0.906** [0.877, 0.935]	0.876 / 0.906	0.675 * / 0.905
28	twcj-j32 / twcj-j16	0.997 [0.954, 1.041]	0.997 / 0.996	0.997 / 0.993
28	twcj-j64 / twcj-j16	0.998 [0.947, 1.053]	0.998 / 0.996	0.998 / 0.986
32	twcj-j16 / rho-neg	**1.366** [1.267, 1.473]	1.244 [1.154, 1.341] / 1.367 [1.267, 1.473]	0.689 [0.639, 0.743] * / 1.368 [1.268, 1.476]
32	f1-x07 / rho-neg	1.595 [1.508, 1.687]	1.504 / 1.595	1.093 * / 1.598
32	twcj-j16 / f1-x07	**0.857** [0.811, 0.905]	0.827 / 0.857	0.631 * / 0.856
32	twcj-j32 / twcj-j16	0.985 [0.930, 1.044]	0.985 / 0.985	0.985 / 0.984
32	twcj-j64 / twcj-j16	0.990 [0.934, 1.049]	0.990 / 0.989	0.990 / 0.986

* The pre-2026-11-08 reading prices rho-neg's per-step canonicalisation and not the walk arms' per-step table inserts. Its ecbench verdict (`data/band-k*.txt`) for rho-neg against twcj-j16, j32 and j64 is "unresolved inside the band" in all 16 sessions each, so no rank is admitted against the twcj arms under that reading; against f1-x07 it is "rho-neg < f1-x07" in 15 of 16 sessions and unresolved in 1 (the f1-x07 inserts are priced at w = 1 through the probe lookups, so its asymmetry is smaller) [pcc:unverified]. Under the priced reading (`--price-table-work`, `data/band-tw-k*.txt`) rho-neg is ahead of every arm at every weight in 16 of 16 sessions [pcc:unverified]. That statement holds under the priced reading only [pcc:unverified].

Phases and walk diagnostics (mean over 512 measured runs per arm, units of sqrt(r))

k	arm	fixed	set-up	walk	collision verify	probes	total	walk steps / sqrt(r) (sd)	collisions per run	set-up GAE
28	rho-neg	-	0.059	0.964	0.006	-	1.029			
28	f1-x07	0.012	0.055 (incl. verify)	0.969		0.548	1.584			
28	twcj-j16	0.012	0.032	1.367	0.024	0	1.435	1.37 (0.48)	2.06	186
28	twcj-j32	0.012	0.044	1.351	0.023	0	1.430	1.35 (0.47)	2.03	255
28	twcj-j64	0.012	0.068	1.329	0.023	0	1.432	1.33 (0.49)	2.01	393
32	rho-neg	-	0.015	0.999	0.002	-	1.015			
32	f1-x07	0.004	0.016 (incl. verify)	1.017		0.582	1.619			
32	twcj-j16	0.004	0.009	1.368	0.007	0	1.387	1.37 (0.52)	2.06	212
32	twcj-j32	0.004	0.012	1.344	0.007	0	1.367	1.34 (0.48)	2.04	290
32	twcj-j64	0.004	0.019	1.343	0.007	0	1.373	1.34 (0.49)	2.01	443

Every twcj record at every jump count solved at rank 2 with zero probe hits on the first trial (3,072 of 3,072) [pcc:unverified]. The set-up is above the frozen model (`data/model.json`): 178.0 GAE at k = 28 and 203.0 at k = 32 for 16 jumps against measured 185.8 and 212.5 (+4.4 %, +4.7 %); the measured set-up cost per extra jump is 4.31 GAE at k = 28 (16 to 32 and 32 to 64 jumps) and 4.82 / 4.79 GAE at k = 32, against 4.5 and 5.0 in the model (the 3.5 to 3.8 GAE of the k = 20 / 24 study does not carry over, and PROTOCOL.md's "about 3.8 GAE per jump" disagrees with its own model.json) [pcc:unverified].

3. Predictions (PROTOCOL.md section 5)

id	prediction	outcome	status

P1	twcj-j16/rho-neg point in [1.30, 1.50] at w = 0, lower end > 1 at w = 0 and (inserts priced) at every weight, both sizes	1.395, 1.366; lower ends 1.345 / 1.347 / 1.356 and 1.267 / 1.267 / 1.268	**held** [pcc:unverified]
P2	twcj-j16/f1-x07 upper end < 1, point in [0.82, 0.93], both sizes	0.906 [0.877, 0.935], 0.857 [0.811, 0.905]	**held** [pcc:unverified]
P3	j16 walk steps in [1.30, 1.42] both sizes; mean S at k = 32 in [1.36, 1.50]	1.367, 1.368; 1.387	**held** [pcc:unverified]
P4	j32/j16 point in [0.95, 1.00] both sizes	0.997, 0.985	**held** [pcc:unverified]
P4b	k = 32 j32/j16 upper end < 1	1.044	**failed** [pcc:unverified]
P5	j64/j16 point in [0.95, 1.01] both sizes	0.998, 0.990	**held** [pcc:unverified]
P6	>= 99.5 % of twcj records at rank 2, no probe, first trial	100 % at every jump count	**held** [pcc:unverified]

4. Reading

- **The distance to rho is flat in k.** With the sibling studies' k = 20 and 24 points (1.358 and 1.343 on the same arm, `research/curve25519_tw_jumps_20261108`), the 16-jump arm is 1.34x to 1.40x rho-neg at w = 0 from k = 20 to 32, and the table-then-probe arm 1.54x to 1.60x [pcc:unverified]. Neither moves toward parity with size.
- **Why 1.37x and not 1.5x.** The derived F = 1 floor ratio 1.5 = 1.329 / 0.886 compares the two walks' model floors; it is not the frozen model, which predicted 1.38 (k = 28) and 1.375 (k = 32) for twcj-j16 / rho-neg, and the data (1.395, 1.366) agree with it within 1.1 % [pcc:unverified]. Measured, the twcj walk runs 3 % above 1.329 at every size, while rho-neg's search runs 9 % to 13 % above 0.886 (0.964 and 0.999 sqrt(r) at k = 28, 32; its look-ahead and cycle escapes are counted) [pcc:unverified]. The ratio of the measured walks is 1.37 to 1.42, and the remaining set-up and verification (0.03 to 0.07 sqrt(r) at these sizes) add little. A better rho-neg walk (closer to its own floor) would widen the gap; nothing here changes the reference.
- **Jumps.** The walk shortens with more jumps (1.367 to 1.344 to 1.343 sqrt(r) at k = 32), by about the amount the 20- and 24-bit study saw, but the gain of 0.02 sqrt(r) is inside the session noise (interval half-width 0.05) and is not established [pcc:unverified]. Sixteen jumps remains the default; 32 is not worse at k >= 28.

5. Distance to matched rho (rho-neg, same sessions)

k	arm	w = 0	w = 0.1 (pre-2026-11-08 / inserts priced)	w = 1 (pre-2026-11-08 / inserts priced)
28	twcj-j16	1.40x [1.35, 1.45]	1.28x / 1.40x	0.72x (artifact) / 1.41x
28	f1-x07	1.54x	1.46x / 1.54x	1.07x / 1.56x
32	twcj-j16	1.37x [1.27, 1.47]	1.24x / 1.37x	0.69x (artifact) / 1.37x
32	twcj-j32	1.35x [1.25, 1.45]	1.23x / 1.35x	0.68x (artifact) / 1.35x
32	f1-x07	1.60x	1.50x / 1.60x	1.09x / 1.60x

Unpriced counters per arm (AGENTS.md section 5a listing)

Mean per measured run, in units of sqrt(r), k = 28 / k = 32 (validator's recomputation from the records) [pcc:unverified]. U is the sum of the counters in `cost.unpriced`; the last column is `work_native.table_insert_attempts`, counted but priced only by the priced reading.

arm	U (cost.unpriced)	components	table inserts (counted, priced only in the priced reading)
rho-neg	0.964 / 0.999	`canonicalisations_uncharged` only	none
f1-x07	0.547 / 0.582	`lookups_uncharged` 0.545 / 0.582, row_ops and legendre_symbols about 0.001	0.969 / 1.017
twcj-j16	0.003 / 0.001	row_ops, legendre_symbols	1.367 / 1.368
twcj-j32	0.003 / 0.001	row_ops, legendre_symbols	1.351 / 1.344
twcj-j64	0.003 / 0.001	row_ops, legendre_symbols	1.329 / 1.343

Each is priced at 0, 0.1 and 1 GAE per operation in the band files, and two arms are ranked only where the order holds across that band [pcc:unverified].

6. Scope and limits

As in the sibling studies: the F = 1 target-in-walk arm is one unknown away from rho (P1 := G makes it a rho walk with a stored table), does no decomposition work, is not decomposition-based index calculus, and proposes no leaderboard or `benchmarks/README.md` row; no VALIDATION.json or replay certificate exists (a validator report now does, `validator-`

report.json) [pcc:unverified]. Single target only. Local L0 counted units, deterministic and replayed; no wall-clock claim. No A/A arm (it would replicate rho-neg exactly). research-kb and ecdb were not reachable; no novelty claim.

Validation (independent validator, 2026-11-08; status pass_with_errata)

Source: validator-report.json in this directory (not edited). Each erratum and how it was applied here [pcc:unverified]:

- **E1** (count): "five of seven" corrected to six of seven held, P4b the only failure, in the Answer; the coverage report carries the same correction [pcc:unverified].
- **E2** (band-verdict footnote): replaced by "unresolved against every twcj arm in all 16 sessions; against f1-x07 rho-neg is ahead in 15 of 16 and unresolved in 1", with the pre-2026-11-08 label on the reading [pcc:unverified].
- **E3** (5a listing): the table of every arm's unpriced counters per run was added to section 5 [pcc:unverified].
- **E4** (set-up numbers and wording): 178 / 203 GAE (model) against 185.8 / 212.5 (measured); 4.3 (k = 28) and 4.8 (k = 32) GAE per extra jump against 4.5 and 5.0 in the model; the 1.37x is compared with the frozen model 1.38 / 1.375 and not with the floor ratio 1.5 [pcc:unverified]. PROTOCOL.md's "about 3.8 GAE per jump" is not edited (frozen) [pcc:unverified].
- **E5** (record hygiene): disclosed below; closed records are not edited and the raw archives are not uploaded by this commit [pcc:unverified].
- Also noted, from check 9: the "distance is flat in k" statement rests on four points (1.358, 1.343, 1.395, 1.366) with a least-squares slope of ln ratio on k of +0.0013 per bit (+1.6 % over 12 bits, inside the intervals); no fit was printed in the original README [pcc:unverified].

Disclosures [pcc:unverified]:

- The measuring binary d23fab99... (/home/user/ctarget-f1/release/ecbench) was rebuilt on disk at 2026-11-08T21:08:01Z during validation (now fcb16d9c...). The validator's replays used fcb16d9c and reproduced every count in all 16 sessions (audit OK: 325 records, 320 replays reproduced, 0 problems), so the replays establish behavioural identity only, not byte identity of source or binary [pcc:unverified].
- The benchrec records 20261108T2051Z-...-k28 and -k32 store 4-decimal metric values that differ from the validator's recomputation by at most 0.04 % (the 3-decimal README values agree), for example k = 28 twcj / rho 1.3951 against 1.39458; the difference comes from rounding in the producer's pipeline. All raw entries have uploaded: false (raw archives not uploaded to S3), and session.json git_commit and git_dirty are null [pcc:unverified].
- For its replays the validator exported CHECKPOINT_SKIP as well as BENCHREC_SKIP; no ledger bypass entry resulted [pcc:unverified].
- Validation gate: validator report only; records have validation: null and no VALIDATION.json exists, so no leaderboard or benchmarks/README.md row follows [pcc:unverified].

Files

PROTOCOL.md, specs/, data/frozen.sha256, data/frozen-at.txt, scripts/ (model.py, make_specs.py, run_sessions.sh, pooled_compare.py, analyse.py, make_figure.py, render_pdf.py), sessions/k*-s*/{session,spec,plan,host}.json (records.jsonl in data/sessions-k{28,32}-records.tar.gz, hashed in the benchmark records), data/ (model, plans, run logs, both band readings, verify reports, pooled_measured.json, binary hashes), figures/scaling.{svg,png}, REPORT.pdf. Benchmark records: benchmarks/records/20261108T2051Z-curve25519-tw-scaling-k{28,32}.json; copies in benchmarks/20261108-curve25519-tw-scaling/.